Pre-Analysis Plan (PAP)

Authors: Ole Brüggemann and Iván Canzio

H1. Having daughters leads to stronger support for gender equality (as measured by attitudes).

Variables:

As there is no direct attitudinal measure of *support for gender inequality*, we will use the item `plj0625` which measures agreement with the following statement "At home, the husband should have the final say" on a 7-point Likert scale. We spend quite some time searching for measures that were asked more frequently and based on the 5% sample of the data we also carried out tests for constructing a broader index for gender equality based on a range of items about the position of women in society (`plj0621,2,3,4`). In the end we decided against it based on our reading of the items and their low correlations. We reverse code the focal item so that higher values stand for higher support for gender equality. Due to the skewed distribution (and the distribution of residuals in tested models), we dichotomize with 1 representing 7 on the previous likert scale ("completely disagree"), and 0 representing values 1 up to 6, with the constructed variable "`d_gender_eq`".

To investigate the effect of *having daughters* we had long discussions about how to construct the treatment variable. In the end we decided to construct a dummy time-varying variable that is 0 when an individual has only boys and 1 when an individual has only daughters. We do this based on the following two reasons: First, identifying the effect of having daughters compared to the reference group of parents with boys and respondents without any children would not allow to disentangle the effect of having daughters from the effect of parenthood. Second, our approach avoids the issue of selection into parenthood which we believe to be crucial to circumvent. We also discussed whether we should only consider the sex of the first child as random or also the subsequent children. Other articles (e.g. Choi et al., 2008; Clarke et al., 2025) only considered the effect of the first child. However, we were not very convinced that measuring the effect of the first child would bias less our results compared to considering the effect of subsequent children too.

The technical approach to create the variable is based on first looping through the children variables (`kidgeb01-19` and `kidsex01-19`) and identifying for each year whether the respondent has a boy and or a girl. And based on these two intermediate variables we generate the treatment dummy of interest (`d_daughter`).

We further create a variable for age (`age`) of respondents by subtracting the birth year from the survey year (`syear`) and a variable for the age squared (`age_sq`) by multiplying the previously generated age variable.

Finally we also use the *sex* of the respondents (`sex`) and the personal id (`pid`) to cluster standard errors at the individual level.

Inclusion/Exclusion criteria:

We restrict our analyses based on the data availability of the variables for certain years – in this hypothesis the dependent variable is only available between 2016 and 2018 – and our coding of the treatment variable restricts the analyses to parents that have either only boys or only daughters. Other control variables we include (age, sex, survey year) are available for all cases.

Exact analysis:

At the beginning we considered implementing a staggered difference-in-differences approach to estimate the effect of having daughters versus having boys or being childless. For this purpose we thought of estimating first the effect boys, then the effect of daughters, and later estimate the difference between the two. However, several issues prevented us from following this approach. First, the number of waves was very small (only three) to perform a staggered DID. Second, we could not have reliably tested the parallel-trends assumption with a 5% sample. Third, we would have needed to calculate the difference between the two effects, meaning that the final estimate is not directly obtained from a regression model. Therefore, our final approach was to simply estimate the effect of having "only daughters" versus having "only sons". To test this effect we execute a linear probability model in which we estimate the effect of having only daughters versus having only boys. As control variables we include individuals' age and its quadratic term, individuals' sex and year dummy variables. We use cluster-robust standard errors to account for repeated individual observations.

Model specification: $GenEqualitySupport_{it} = \beta_0 + \beta_1 OnlyDaughters_{it} + \gamma X_{it} + \varepsilon_{it}$

$GenEqualitySupport_{it}$ is the dependent variable that represents support for gender equality for the individual i at time t.

β_0 is the intercept.

$OnlyDaughters_{it}$ is a dummy variable indicating if respondents have "only daughters" or "only boys" for individual i at time t and β_1 is its coefficient.

X_{it} is a group of control variables (sex, age, age squared, and dummy survey years) and γ is a column vector representing the coefficient of these control variables.

ε_{it} represents the error term for each individual and time t.

Interpretation of results:

We focus on the regression coefficient for our treatment variable of having only boys versus only having boys. A coefficient that is greater than zero and has a p-value smaller than 0.05 is interpreted as support for the hypothesis. In our small 5% sample, with only 103 observations, we find that the effect of having "only daughters" (compared to having "only boys") is negative (β = -0.0119) and not significant (p. = 0.905).

H2. Having daughters leads to more left-leaning political preferences.

Variables:

We measure *left-leaning political preference* based on the general party preference (`plh0011_h`, `plh0012_h`) as these variables are available for each year while the frequently used left-right scale is only asked every four to five years. We construct a dummy variable (`left`) where respondents are coded as left-leaning when they have a general party preference for parties that are left of the center of the political spectrum (SPD; Grüne; Linke). Having no general party preference as well as preferences for all the other parties are coded as zero.

Our treatment variable for *having daughters* (variable `d_daughter`) is constructed based on the approach described for the first hypothesis.

As described in the first hypothesis, we further create a variable for *age* of respondents by subtracting the birth year from the survey year (`age`) and a variable for the age squared by multiplying the previously generated age variable (`age_sq`).

Finally we also use the *sex* of the respondents (`sex`) and the personal id (`pid`) to cluster standard errors at the individual level.

Inclusion/Exclusion criteria:

We restrict our analyses based on the coding of the treatment variable which reduces the analyses to parents that have either only boys or only daughters. Finally, our analytical sample is reduced to observations that are non-missing for all the variables we use in the regression model.

For this hypothesis this means that we are using data from 2000 until 2020. Other control variables included in the model – age, year, sex – are available for all observations.

Exact analysis:

To test the second hypothesis, we specify a linear probability model with year fixed effects and cluster-robust standard errors for the nesting of years within individuals.

Model specification: $LeftVote_{it} = \beta_0 + \beta_1 OnlyDaughters_{it} + \gamma X_{it} + \varepsilon_{it}$

$LeftVote_{it}$ is the dependent variable that represents left-leaning political preferences for individual *i* at time *t*.

β_0 is the intercept.

$OnlyDaughters_{it}$ is a dummy variable indicating if respondents have “only daughters” or “only boys” for individual *i* at time *t* and β_1 is its coefficient.

X_{it} is a group of control variables (sex, age, age squared, and dummy survey years) and γ is a column vector representing the coefficient of these control variables.

ε_{it} represents the error term for each individual and time t.

Interpretation of results:

We focus on the regression coefficient for our treatment variable of having only boys versus only having boys. A coefficient that is greater than zero and has a p-value smaller than 0.05 is interpreted as support for the hypothesis. We find that the effect of having a daughter on support for leftist parties is positive (*β = 0.0403*) but insignificant at conventional levels (p. = 0.083).

H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

Variables:

"Gender equality in the household" is a very broad concept. Since we are required to produce a single regression output, we decided not to include gender equality with regard to employment due to its complexity. We opted for measuring gender equality with respect to housework division (variable `pli0043_h`) since this indicator is available for most waves and housework inequality is a crucial dimension of gender inequality within households. We had some discussions about what exactly meant gender equality. Intuitively, we would have assumed that more gender equality could be represented by a higher housework contribution by men, since most women perform more housework than their male partners. However, we decided to interpret the hypothesis literally and assume that gender equality represented more symmetry in housework division between two partners of different gender. Therefore, we constructed a variable that captures absolute differences in housework hours between the male and the female partner, where values corresponding with zero represent absolute equality (this is, men and women do the same housework hours) and positive values represent deviations from absolute equality. To construct this variable we simply addedto the yearly observations for males the housework values of their female partners as long as these values ranged between 0 and 99 hours. Later, we calculated the absolute difference (variable `hh_gender_equal`). We top code the dependent variable as a maximum of 10 hours of difference, which affected less than 1% of the values.

Our treatment variable for *having daughters* is constructed based on the approach described for the first hypothesis.

We further create a variable for *age* of respondents by subtracting the birth year from the survey year and a variable for the age squared by multiplying the previously generated age

variable. As our model is at the household level, we create a variable for the female's partner's age and age squared as well (`wife_age`; `wife_age_sq`).

Finally we use the personal id (`pid`) to cluster standard errors at the individual level.

Inclusion/Exclusion criteria:

Since the analyses consisted of couples, we restricted the analyses to observable couples (therefore excluding single individuals or individuals whose partners cannot be observed) in which none of the two members presents missing values for the dependent or independent variables. Additionally, we restricted the sample to those couples that share the same gender composition of their children to ensure that both partners are exposed to similar treatments. This means, for example, that we excluded partners who had a child in a previous relationship if their child does not match in gender with the child of their partner, but these re-partnered individuals are included if both have a similar composition of children (e.g. both have only boys or both have only daughters).

Exact analysis:

We estimate a linear regression model in which the dependent variable is the absolute difference in hours of housework between men and women, the independent variable is a dummy variable indicating if partners have "only daughters" or "only sons". As control variables we include age of the male partner and its quadratic term, age of the female partner and its quadratic term, and year. Since we have repeated observations of couples, we cluster the standard errors at the couple level. More precisely, we only analyse the observations of the male partners, to which we attribute values of the female partners for the dependent and control variables.

$$HouseworkDifference_{it} = \beta_0 + \beta_1 OnlyDaughters_{it} + \gamma X_{it} + \varepsilon_{it}$$

$HouseworkDifference_{it}$ is the dependent variable that represents male-female absolute differences in housework hours for couple i at time t.

β_0 is the intercept.

$OnlyDaughters_{it}$ is a dummy variable indicating if the couple has"only daughters" or "only boys" for couple i at time t and β_1 is its coefficient.

X_{it} is a group of control variables (age of the male partner, age squared of the male partner squared, age of the female partner, age of the female partner squared and dummy survey years) and γ is a column vector representing the coefficient of these control variables.

ε_{it} represents the error term for each couple and time t.

Interpretation of results:

We focus on the regression coefficient for our treatment variable of having only boys versus only having boys. A coefficient that is greater than zero and has a p-value smaller than 0.05 is interpreted as support for the hypothesis. We find that the effect of having “only daughters” (compared to having “only boys”) is positive (β = 0.134) but not significant (p. = 0.265).

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

Variables:

Since the outcome is the same as in the first hypothesis, we did the same selection of variables. To capture "support for gender equality", we used the item `plj0625` measuring agreement with the statement “At home, the husband should have the final say” with a 7-point Likert scale. Again, due to the skewed distribution, we dichotomized the item as "completely disagree" (value 1) or else (value 0) in the variable `d_gender_eq`.

As treatment, we used again a dummy variable indicating if participants have only boys (value 0) or only daughters (only 1) for each of the survey years (variable `d_daughter`). We constructed this variable from variables `kidgeb01-19` and `kidsex01-19`. This means that the independent variable will allow us to compare the effect of having daughters versus having boys, and not the effect of having daughters versus not having daughters (which we cannot identify).

We further create a variable for *age* of respondents by subtracting the birth year from the survey year (`syear`) and a variable for the age squared (`age_sq`) by multiplying the previously generated age variable.

Finally we also use the *sex* of the respondents (`sex`) and the personal id (`pid`) to cluster standard errors at the individual level.

Inclusion/Exclusion criteria:

We follow the same strategy we followed for hypothesis 1.

Exact analysis:

We estimate the effect of having “only daughters” versus having “only sons”, conditional on the gender of the respondent. To test this effect we execute a linear probability model in which we estimate the effect of having only daughters versus having only boys and interact it with the gender of the respondent. As control variables we include individuals’ age and its quadratic term, individuals’ sex and year dummy variables. We also control for the effect of the control variables conditional on gender and conditional on having daughters by

interacting the independent variable and the moderator with each of the control variables. We use cluster-robust standard errors to account for repeated individual observations.

$GenEqualitySupport_{it} = \beta_0 + \beta_1 OnlyDaughters_{it} + \beta_2 Sex_i + \beta_3(OnlyDaughters_{it} \cdot Sex_i) + \delta(OnlyDaugthers_{it} \cdot X_{it}) + \zeta\,(Sex_i \cdot X_{it})$

$GenEqualitySupport_{it}$ is the dependent variable that represents support for gender equality for the individual *i* at time *t*.

β_0 *is the intercept*

$OnlyDaughters_{it}$ is a dummy variable indicating if respondents have “only daughters” or “only boys” for individual *i* at time *t* and β_1 is its coefficient.

X_{it} is a group of control variables (sex, age, age squared, and dummy survey years) and γ is a column vector representing the coefficient of these control variables.

δ is a column vector that represents the coefficients of the interactions between the independent variable *OnlyDaugthers* and the group of control variables *X.*

ζ is a column vector that represents the coefficients of the interactions between the independent variable *Sex* and the group of control variables *X.*

ε_{it} represents the error term for each individual and time t.

Interpretation of results:

We focus on the regression coefficient of the interaction of our treatment variable with gender. A coefficient that is greater than zero and has a p-value smaller than 0.05 is interpreted as support for the hypothesis. We find that the effect of having “only daughters” (compared to having “only boys”) conditional on gender is positive (β = 0.453) but not significant (p. = 0.848).